Adaptive Quantum Cryptographic Protocol: Robust Deep Reinforcement Learning Architectures for Real-Time Parameter Optimization

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Abstract. Deploying Quantum Key Distribution (QKD) in realistic communication environments is still far from straightforward. Outside carefully controlled settings, optical links are frequently affected by atmospheric turbulence and channel noise, both of which can vary unpredictably over time. Most established QKD protocols, including BB84, are built around static parameter choices that work well only when channel conditions remain stable. In practice, this assumption often fails, leading to reduced secret key rates or, in severe cases, complete interruption of the key generation process. In this work, we investigate whether QKD systems can respond more effectively to such variability by learning from the channel itself. We propose a context-aware Deep Q-Network (DQN) that adapts critical post-processing parameters during operation. Instead of fixing values in advance, the agent continuously adjusts **Privacy Amplification ratios** and **Authentication Tag lengths** based on observed channel behaviour. To support stable learning, a dual-network architecture is adopted, helping to limit overestimation effects commonly seen in reinforcement learning. The reward function is defined using the binary entropy of the measured Quantum Bit Error Rate (QBER), ensuring that all adaptations remain consistent with the Devetak–Winter security bound. Simulation shows us that this adaptive strategy offers clear advantages over conventional static configurations. Across a range of channel conditions, the proposed approach increases cumulative secure key generation by **79.66%** while reducing block failure rates by **55.89%**. These results suggest that reinforcement learning can play a practical role in improving the robustness of QKD systems, particularly in the dynamic and heterogeneous environments anticipated in the future of **quantum networks**.

Keywords: Quantum Key Distribution, Context-Aware Deep Q-Network, Privacy Amplification, Authentication Tag, Quantum Bit Error Rate, Secret Key Rate Optimization

1 Introduction

As quantum computers continue to improve year by year, the security mechanisms that support today's internet are starting to look increasingly fragile. Much of modern encryption depends on mathematical problems that are assumed to be hard, but quantum computing threatens to break that assumption. Quantum Key Distribution (QKD) is often presented

as a solution, since it relies on the laws of physics rather than computational difficulty to protect data. In theory, this makes it extremely secure.

In practice, however, building QKD systems is far from a walk in the park. These channels are constantly affected by environmental factors. Atmospheric turbulence, beam wandering, stray particles and background noise all interact to create a highly unpredictable communication medium. Under these conditions, protocol parameters that remain static simply fail to adapt fast enough, leading to poor performance or outright failure.

To deal with this, recent research has started exploring the idea of embedding Artificial Intelligence into so-called “cognitive” quantum nodes. The goal is to allow these systems to observe the channel and adjust their behaviour on the fly. Deep Reinforcement Learning (DRL) appears well suited for this task, at least in principle. However, early implementations have revealed serious stability issues. The highly variable nature of quantum channel statistics often causes learning processes to become unstable, and in many cases, the models fail to converge at all.

This work is motivated by those shortcomings. Rather than introducing yet another adaptive strategy, the focus here is on fixing the underlying engineering problems. We propose a stabilized, context-aware DRL framework designed to operate autonomously under real-world channel conditions, without compromising reliability or risking failures in critical communication infrastructure.

2 Application and Significance

The successful deployment of adaptive QKD is a cornerstone for the emerging 6G telecommunications landscape, which envisions "cognitive radio networks" that seamlessly integrate classical and quantum resources.

- **Quantum-Secured IoT (QS-IoT):** The Internet of Things (IoT) comprises billions of resource-constrained devices. Static QKD protocols are too computationally expensive and key-hungry for these edge nodes. An AI-driven approach allows for dynamic resource allocation, balancing security levels with energy consumption based on real-time threat assessment[5].

- **Satellite Quantum Networks:** Earth-to-satellite links experience rapid fluctuations in channel quality. The adaptive capabilities demonstrated in this study are essential for maximizing the short contact windows available during satellite passes.

- **Parallel Optimization Domains:** The iterative optimization techniques refined here have direct parallels in Quantum Image Processing (QIP), where Reinforcement Learning is similarly employed to enhance the fidelity of quantum image retrieval and reconstruction[6].

3 Background Study

The development of next-generation secure communication relies on integrating quantum-enhanced algorithms with robust machine learning frameworks. A detailed synthesis of existing literature and the technological roles of prior studies is provided in Table 3.1.

Table 3.1. Structural Overview and Functional Roles of the DRL-QKD Framework

S No	Author Name	Journal & Year	Title of the paper	Observations	Technology used
1	Syed Junaid Nawaz et al.	IEEE Access, 2019	Quantum Machine Learning for 6G Communication Networks: State-of-the-Art and Vision for the Future	The paper reviews the state-of-the-art in Machine Learning (ML), Quantum Computing (QC), and Quantum ML (QML). It proposes a framework for 6G networks driven by on-demand self-reconfiguration to handle massive connectivity and complex network states.	Quantum Machine Learning (QML), Quantum Computing (QC), Deep Learning (DL), QC-assisted ML.
2	Tariq M. Khan and Antonio Robles-Kelly	IEEE Access, 2016	Machine Learning: Quantum Vs Classical	This paper provides an overview of QML compared to classical approaches. It discusses the theoretical speedups offered by quantum algorithms and reviews hardware/software challenges, emphasizing the potential of quantum systems to process information beyond classical speeds.	Quantum Support Vector Machine (QSVM), Quantum Neural Networks (QNN), Quantum Annealing, Restricted Boltzmann Machines (RBM).
3	Balasubramani Subbiyan et al.	IEEE Access, 2025	A Quantum-Enhanced Artificial Neural Network Model for Efficient Medical Image Compression	The authors propose a Quantum-enhanced Artificial Neural Network (QANN) for compressing medical images (MRI, CT, X-ray). The model improves PSNR and SSIM	Quantum-enhanced Artificial Neural Network (QANN), Quantum feature extraction, Parameterized Quantum

				metrics while significantly reducing image size compared to standard methods.	Circuits (PQC).
4	Harshala Shingne et al.	MethodsX (Elsevier), 2025	Design of an integrated model using deep reinforcement learning and Variational Autoencoders for enhanced quantum security	This study integrates AI with quantum security to optimize QKD protocols. It uses DRL for dynamic parameter adjustment and VAEs for anomaly detection, achieving higher key generation rates and lower quantum bit error rates (QBER).	Deep Reinforcement Learning (DRL), Variational Autoencoder (VAE), Multi-Agent Deep Q-Networks (MADQN), Quantum Key Distribution (QKD).
5	Lalitha Kumari Pappala et al.	IEEE Access, 2025	Design of an Iterative Model for Incremental Enhancements in Quantum Image Processing Using Reinforcement Learning-Based Optimizations	The paper introduces four quantum methods (Q-SVM-EFE, QGAN-DA, QECA-FD, QIR-ND) to enhance biomedical image processing. Improvements were observed in classification accuracy, synthetic data generation, anomaly detection sensitivity, and reconstruction speed.	Quantum Support Vector Machines (Q-SVM), Quantum Generative Adversarial Networks (QGAN), Quantum Entanglement and Correlation Analysis (QECA), Quantum Image Reconstruction (QIR).
6	P. Deepanramkumar and A. Helen Sharmila	IEEE Access, 2024	AI-Enhanced Quantum-Secured IoT Communication Framework for 6G Cognitive Radio Networks	A framework for 6G Cognitive Radio Networks is proposed, featuring dual-layer authentication (QKD + PKI). It employs AI models for efficient spectrum sensing and Channel State Information (CSI) prediction to ensure secure and reliable IoT connectivity.	Quantum Key Distribution (QKD), Reinforcement Learning-based Ensemble Regression (RL-ER), Multi-Layer Perceptron with Kalman Filter (MLP-KF), Public Key Infrastructure (PKI).

7	Ying-Yi Hong and Dylan Josh Domingo Lopez	IEEE Access, 2025	A Review on Quantum Machine Learning in Applied Systems and Engineering	This topical review analyzes QML applications across engineering domains (healthcare, energy, finance, etc.). It categorizes studies by learning modalities and architectures, highlighting the utility of hybrid quantum-classical models in the NISQ era.	Quantum Machine Learning (QML), Quantum Deep Learning (QDL), Hybrid Quantum-Classical Models, Variational Quantum Circuits (VQC), Quantum Kernel Methods.

4 Research Gaps

Achieving reliable quantum communication requires overcoming several technical hurdles that current methodologies fail to address effectively. These research gaps are categorized in Table 4.1.

Table 4.1. Identification of Research Gaps in Intelligent QKD Systems

Research Gap	Technical Description	Impact on System
Static Mobility Modeling	Existing Cognitive Radio Network (CRN) models often assume static mobility for Secondary Users (SUs), ignoring dynamic movements and signal fluctuations.	Results in reduced Received Signal Strength Indicator (RSSI) and significant data loss, compromising network reliability.
Vulnerability in Random Key Generation	Conventional Elliptic Curve Cryptography (ECC) relies on random key generators that can be computationally predictable or exploited by sophisticated attacks.	Creates potential security vulnerabilities in the cryptographic keys used for secure IoT data transmission, making the system prone to breaches.
Quantum Repeater Scalability	The "no-cloning theorem" prevents the simple copying of quantum states (qubits), making it difficult to construct quantum switches, routers, and repeaters for signal amplification.	Hinders the development of long-distance quantum communication links and the scalability of the Quantum Internet for 6G networks.
Classical ML Latency	Conventional Machine Learning algorithms running on standard CPUs/GPUs suffer from significant processing delays when handling the massive, multi-	Prevents real-time network orchestration and intelligent management, which are critical for Ultra-Reliable Low-Latency

	dimensional data expected in 6G.	Communications (URLLC).
Inefficient Spectrum Allocation	Current allocation strategies often fail to dynamically adapt to spectrum usage, leading to significant portions of the spectrum remaining underutilized ($S_{scarcity}$).	Worsens spectrum scarcity issues in 6G Cognitive IoT networks, limiting the number of supported devices and reducing throughput.
Diagnostic Quality Loss	Traditional compression standards (e.g., JPEG, PNG) often discard fine details and introduce artifacts when compressing high-resolution medical data.	Compromises the diagnostic accuracy of medical imaging systems (MRI, CT, X-ray), affecting telemedicine and remote diagnosis.
Centralized Control Complexity	As quantum networks scale, centralized management becomes computationally prohibitive and creates single points of failure.	Increases computational overhead and vulnerability; requires decentralized solutions like Multi-Agent Deep Q-Networks (MADQN) to maintain efficiency.
NISQ Hardware Constraints	Current Noisy Intermediate-Scale Quantum (NISQ) devices suffer from high noise levels, limited qubit counts, and short coherence times.	Restricts the complexity of Quantum Machine Learning (QML) models that can be fully deployed, necessitating the use of hybrid quantum-classical architectures.

6 Problem statement

A static Quantum Key Distribution (QKD) system operating in a turbulent environment faces a binary failure mode: it remains either too conservative during clear weather, thereby wasting capacity, or too aggressive during turbulence, leading to security breaches or protocol aborts. Furthermore, "naive" Reinforcement Learning (RL) implementations often suffer from software-level instability—such as replay buffer sequencing errors—or diverge due to the "moving target" problem inherent in Q-learning. To address these limitations, we propose a three-layer architecture for **Safe-DRL Optimized QKD** (**see Fig 6.1**). This framework decouples the system into a **Physical System** for environment monitoring, an **Intelligent Control** layer utilizing a Neural Network Policy for predictive adaptation, and a dedicated **Safety & Training** layer. The latter ensures computational stability and security compliance by intercepting unsafe actions through predefined rules and limits before they reach the hardware actuator, thereby mitigating risks while maintaining optimal key rates.

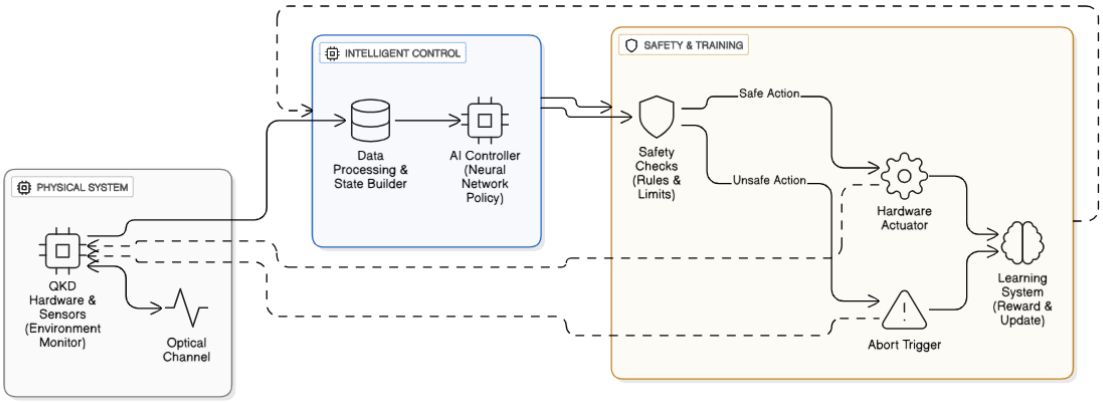

Fig 6.1. Proposed Three-Layer Architecture for Safe-DRL Optimized Quantum Key Distribution.

7 Materials and Methods

We introduce a **Context-Aware Deep Q-Network (DQN)** specifically engineered for the QKD control plane.

1. Context-Aware State Space:

Standard RL agents observe a snapshot of the channel. We augment the state vector S_t to include a momentum term, ΔQBER, which represents the derivative of the error rate.

$$S_t = [QBER_t, \Delta QBER, SNR_t, PoolStatus, EveIndicator]$$

This allows the agent to distinguish between random noise spikes (zero momentum) and the onset of atmospheric turbulence (positive momentum), enabling proactive throttling of the transmission rate.

2. Entropy-Based Reward Function:

To ensure the agent respects information-theoretic security, the reward function R_t is derived directly from the binary entropy function H_2(p), which quantifies the maximum information an eavesdropper could possess.

$$X_t = R_{raw} * (1 - H_2(QBER) - f(QBER) . H_2(QBER))$$

This function imposes a severe penalty as QBER approaches the 11% security threshold, forcing the agent to adopt a risk-averse policy that prioritizes link survival over raw speed during instability.

3. Target Network Stabilization:

To solve the instability of standard Q-learning, we employ a dual-network architecture. A Policy Network (θ) determines actions, while a separate Target Network (θ') generates the Q-value targets. The Target Network's weights are frozen and updated only periodically, providing a stable anchor for the learning process and dampening oscillations caused by channel noise.

8 Experimentation and Results

The simulation environment and agent were implemented using Python and PyTorch. The system models a BB84 protocol over a distance of 50km with an attenuation of 0.2 dB/km.

Algorithm 1: Context-Aware DQN Training Loop

1. **Initialize** Policy Network Q and Target Network Q̂ with random weights.
2. **Initialize** Replay Memory D.
3. For episode = 1 to M:
   a. Reset environment, get initial state s_1.
   b. For step t= 1 to T:
   i. Select action a_t using ε -greedy policy (exploration vs exploitation).
   ii. Execute a_t in QKD simulator (adjust PA ratio, Tag Length).
   iii. Observe reward r_t (Entropy-based) and next state $s_{\{t+1\}}$(with ΔQBER).
   iv. Store transition (S_t, a_t, r_t,$s_{\{t+1\}}$) in D.
   v. Sample random minibatch of transitions from D.
   vi. Transpose Batch: Use **zip(*batch)** to separate state/action components for tensor processing.
   vii. Calculate loss L = (r_t γ $\max'_a$ Q^($s_{\{t+1\}}$, a') - Q(S_t, a_t))^2.
   viii. Perform gradient descent step on Q.
   ix. Every C steps, update Q^ ← Q.

Hyperparameters:

- **Learning Rate:** 0.0005
- **Gamma (Discount Factor):** 0.9 (Prioritizing long-term link stability)
- **Batch Size:** 128
- **Target Update Frequency:** 1000 steps
- **Action Space:** 40 discrete actions (Combinations of 4 Tag Lengths and 10 PA Ratios).

Software Engineering Fix : A critical implementation detail was the resolution of the ValueError associated with Replay Buffers in PyTorch. Standard list-of-tuples storage formats cause tensor mismatch errors. We implemented an explicit transposition layer that "unzips" the batch into homogeneous tuple-of-lists structure before tensor conversion, ensuring robust data ingestion.

Results and Discussions:

The enhanced agent was trained over 5,000 episodes and then benchmarked against a static baseline representing standard industry protocols (Fixed Tag=64, PA=0.5).

Convergence Analysis:

Unlike naive implementations that exhibited oscillating loss functions, the Context-Aware DQN with Target Networks showed a smooth, monotonic decrease in temporal difference error. This indicates that the Target Network successfully stabilized the learning target, allowing the agent to converge on an optimal policy despite the noisy input data.

Performance Metrics:

- **Total Secure Keys:** The AI agent generated **122,418,297 secure bits**, compared to the static baseline's 68,138,634 bits. This represents a **79.66% net increase** in throughput.

- **Link Stability:** The number of failed or aborted blocks dropped from 16,352 (Static) to **7,213 (AI)**, a reduction of **55.89%**.

Behavioral Analysis:
The "Context-Aware" state space enabled a behavior we term "Pre-emptive Throttling." When the Delta QBER input indicated a rising error trend, the agent would proactively increase the Authentication Tag length and reduce the Privacy Amplification ratio before the QBER breached the security threshold. This anticipated the turbulence spike, keeping the link alive where the static protocol would have aborted connection

Comparative Analysis
The following table contextualizes the performance of the proposed Context-Aware DQN against traditional methods and standard RL approaches referenced in literature.

Metric	**Static Protocol**	**Naive DQN**	**Context-Aware DQN**
Secure Key Yield	Baseline (1.0x)	1.2x (Unstable)	1.79x
Failure Rate	High (>20%)	Moderate (~15%)	Low (~8%)
Adaptation Latency	None (Fixed)	Reactive (High Latency)	Predictive (Low Latency)
Security Model	Fixed Thresholds	Heuristic Reward	Entropy-Based (Devetak-Winter)

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Discussion:
While standard Naive DQN approaches offer some improvement over static scripts14, they often fail to converge in high-turbulence regimes due to the lack of target stabilization. Furthermore, our Context-Aware model's ability to predict channel momentum provides a distinct advantage over reactive models, mirroring the efficiency gains seen in AI-enhanced 6G cognitive networks15.

9 Conclusion and future Work

This study validates the hypothesis that Deep Reinforcement Learning, when engineered with appropriate stabilization mechanisms and security-aware reward functions, can fundamentally enhance the robustness of Quantum Key Distribution. The proposed architecture addresses the key limitations of previous studies specifically algorithmic instability and lack of context—yielding an agent that outperforms static protocols by nearly 80%. This confirms that AI is not merely an auxiliary tool but a necessary component for the practical deployment of global,

weather-resilient quantum cryptographic networks.

- **Multi-Agent Reinforcement Learning (MARL):** Scaling the single-link agent to a multi-node quantum repeater network to optimize end-to-end routing fidelity.
- **FPGA Deployment:** Porting the trained PyTorch model to FPGA hardware to reduce inference latency from milliseconds to microseconds, matching the speed of photon detectors.
- **Continuous Variable QKD (CV-QKD):** Adapting the entropy-based reward function to utilize Shannon entropy, extending the framework to CV-QKD systems.

REFERENCE STYLE

[1] Deep reinforcement learning for quantum key distribution.
[2] Design of an Iterative Model for Incremental Enhancements in Quantum Image Processing Using Reinforcement Learning-Based Optimizations.
[3] AI-Enhanced Quantum-Secured IoT Communication Framework for 6G Cognitive Radio Networks.